



AN EMPIRICAL STUDY ON RATING AND TAG PREDICTION FOR CODEFORCES PROBLEMS



15TH JUNE, 2026
MONDAY

CSE 4622

Problem Statement

How can a machine learning system automatically predict the difficulty rating and algorithmic topic tags of a competitive programming problem given only its textual statement and structural metadata?

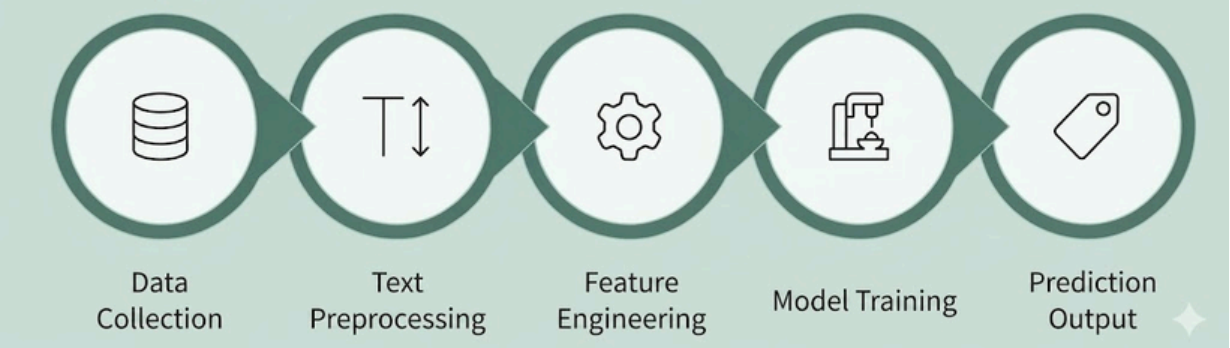
Background

- Codeforces problem ratings & tags initially estimated by experts, later refined using large-scale community solving behavior and contest statistics.
- Final ratings reflect crowd-based perceptions of algorithmic difficulty, making them useful for studying human problem-solving complexity.
- Competitive programming problems contain rich textual, mathematical, and structural information that can be analyzed using machine learning techniques.

Conventional pipeline:

Human Expert → Manual Rating → Manual Tagging → Published Problem

Our pipeline



Formal Definition

Given a problem instance:

$$d = (\text{statement}, \text{time_limit}, \text{memory_limit}, \text{metadata})$$

The objective is to learn two functions simultaneously:

Task A — Tag Prediction (Multi-Label Classification):

$$f_A : d \rightarrow y \in \{0, 1\}^L$$

where L is the number of unique tag labels and $y_l = 1$ indicates the l -th tag applies.

Task B — Rating Prediction (Regression):

$$f_B : d \rightarrow r \in \mathbb{R}$$

where r is the continuous difficulty rating in the range $[800, 3500]$.

□ **Problem Statement (excerpt):**

"You are given an array of n integers. Find the minimum number of operations to make all elements equal, where each operation increments any element by 1. Output the answer modulo $10^9 + 7$."

Ground Truth:

Rating: **1600**

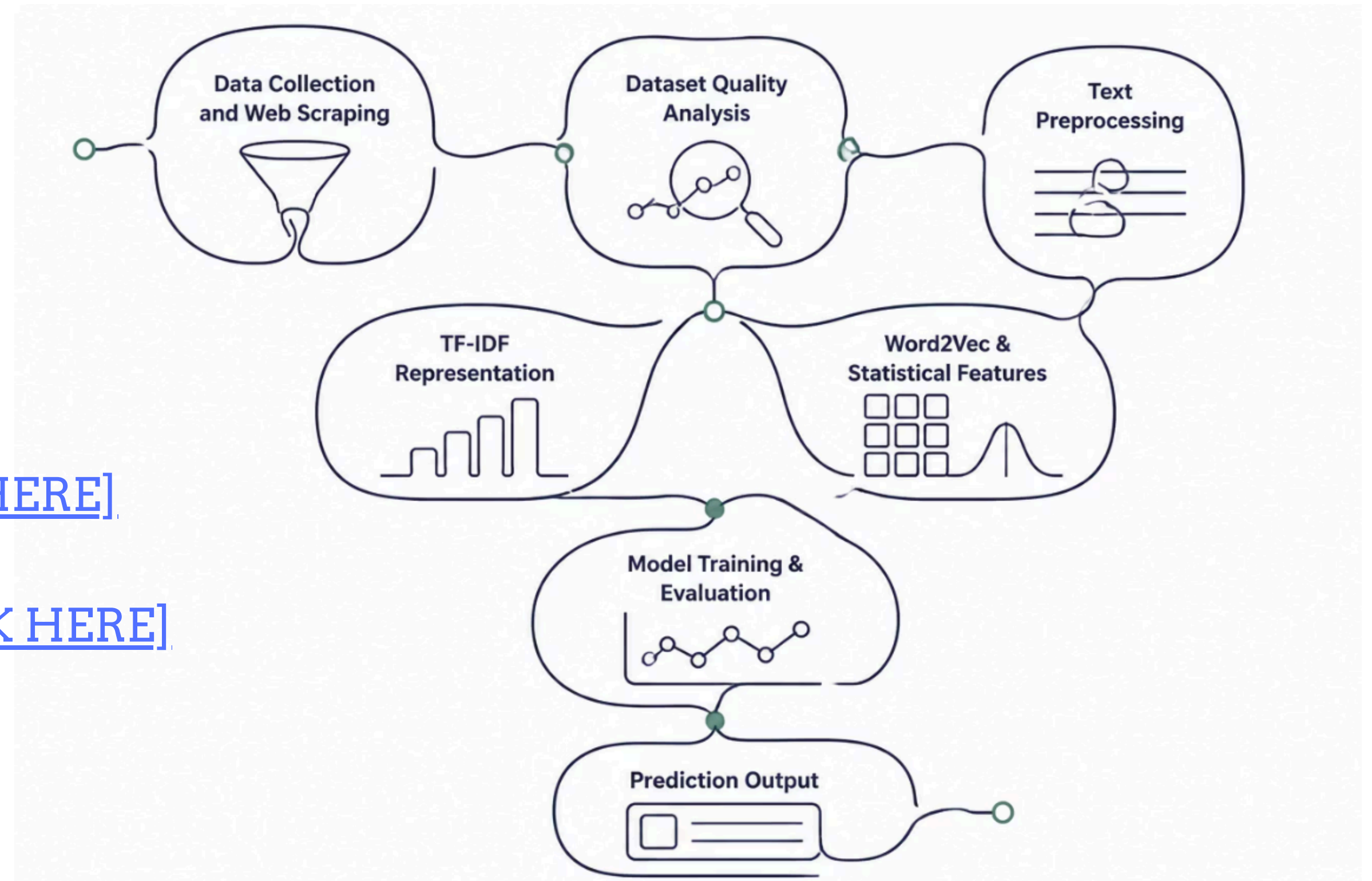
Tags: **[math, greedy, number theory]**

The core challenge is to transform d into a **generalized feature representation**:

$$x = \varphi(d)$$

that captures both semantic content and structural properties of the problem.

Methodology



[SIMPLE DIAGRAM \[CLICK HERE\]](#)

[DETAILED DIAGRAM \[CLICK HERE\]](#)

Dataset

The dataset $\mathcal{D}$ is formally defined as:

$$\mathcal{D} = \{(d_i, r_i, T_i)\}_{i=1}^N$$

where d_i is the i -th problem instance, $r_i \in \mathbb{R}$ is the difficulty rating label, $T_i \subseteq \Lambda$ is the set of applicable tags from label space Λ , and $N = 11,156$ is the total number of problem instances.

Dataset Summary Statistics

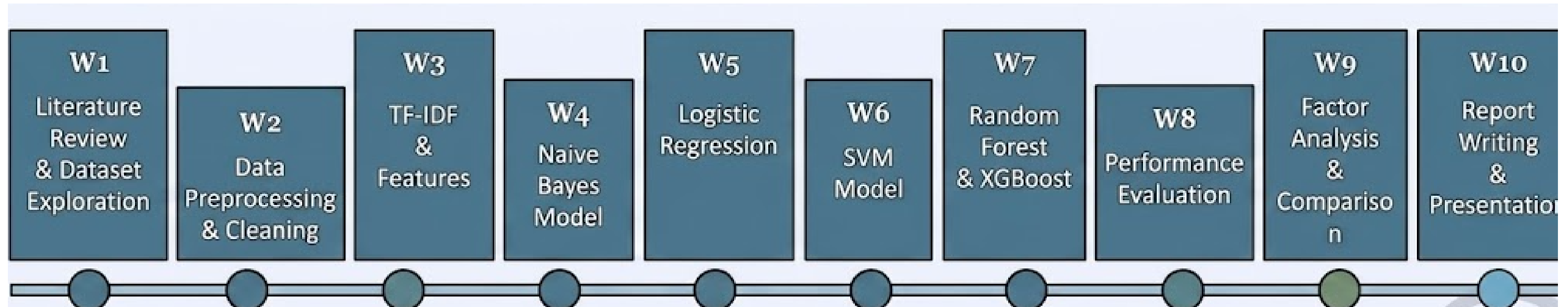
- Total problems: **N = 11,156**
- Total attributes per problem: **17**
- Unique tag categories: **35+**
- Rating range: **[800, 3500]**, step size 100
- Average tags per problem: **2.3**; Maximum: **8**
- Problems with rating: **10,886**
- Problems with tags: **11,011**

- All 11,156 rows contain the correct number of columns – no broken, malformed, or shifted rows detected
- Each instance contains 17 attributes
- Data Preprocessing was done via
 - removal of HTML tags and special characters, lowercasing, tokenization,
 - stop-word removal, lemmatization, removal of duplicate or incomplete entries

Tools & Framework

 PROGRAMMING & ACQUISITION ● Python ● Codeforces API ● BeautifulSoup ● Requests	 LIBRARIES ● Pandas & NumPy ● Scikit-learn ● XGBoost & LightGBM ● Gensim	 FEATURE ENGINEERING ● TF-IDF (Unigram/Bigram) ● Word2Vec ● Statistical Metadata
 VISUALIZATION ● Matplotlib ● Seaborn	 ENVIRONMENT ● Google Colab ● Jupyter Notebook	 VERSION CONTROL ● GitHub

Project Timeline

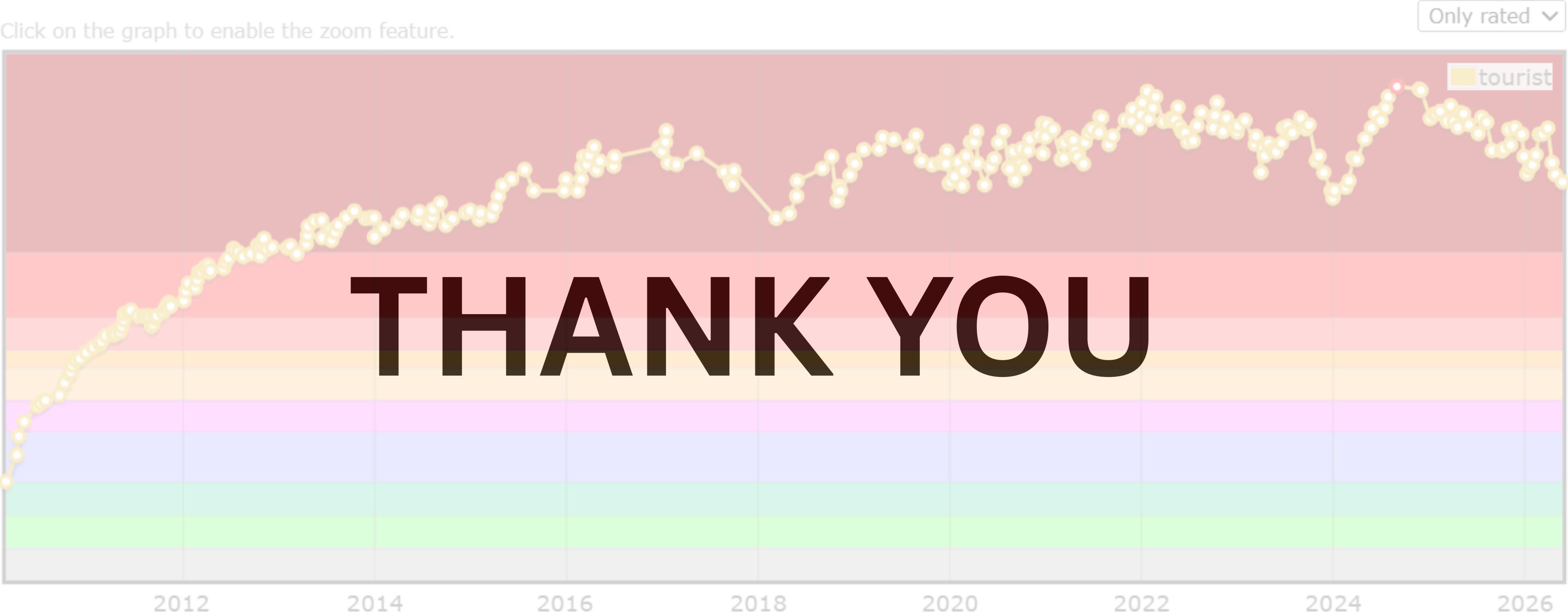


TEAM MEMBERS

Farhan Ishraq
220041217

Arifin Rafi
220041230

Tashin Mustakim
220041239



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